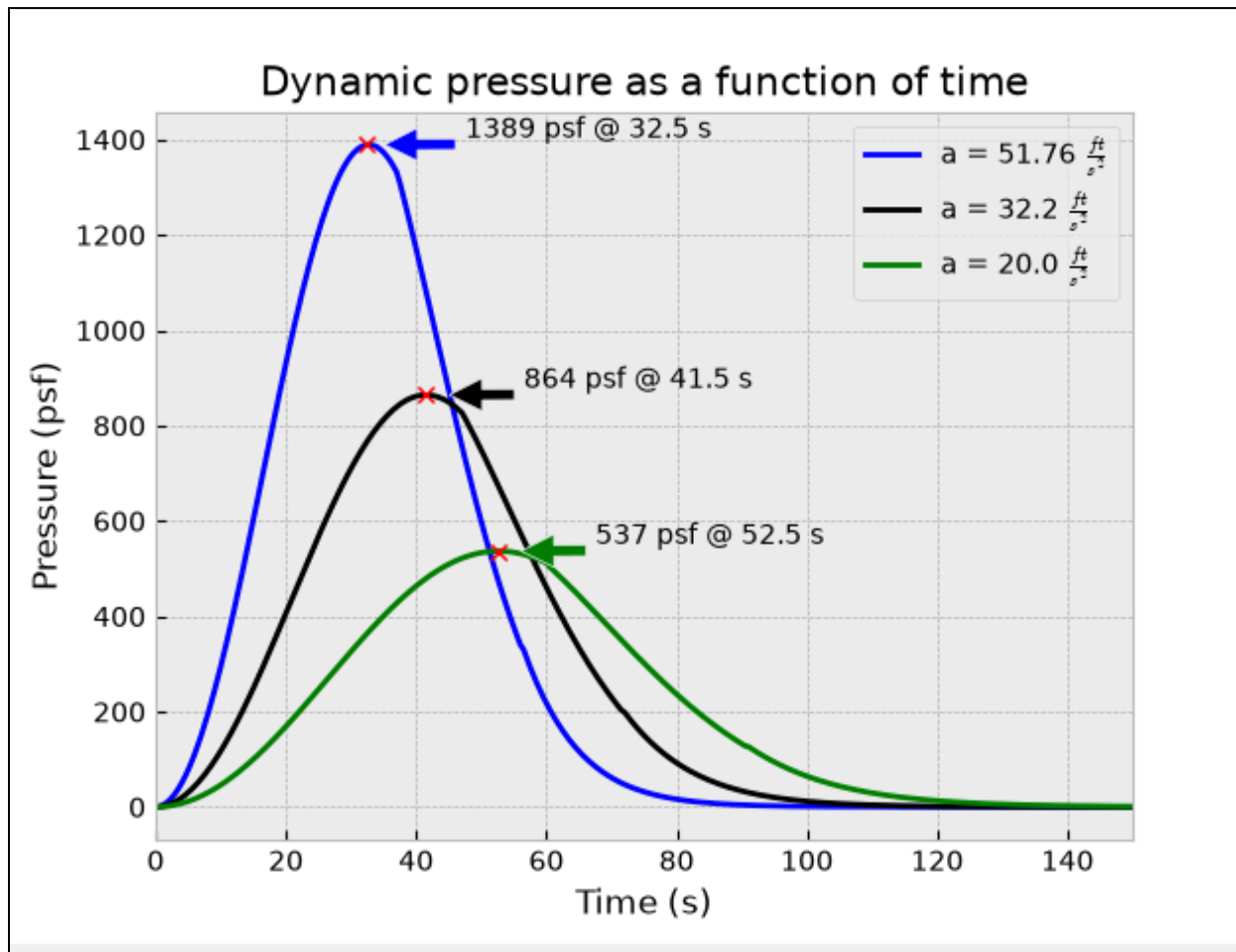


chapter 3 - dynamic pressure



This Python script calculates and visualizes the dynamic pressure experienced by a rocket over time during launch under three different constant acceleration profiles. It computes altitude and velocity using basic kinematics, then applies a segmented atmospheric model to determine air density and calculate the resulting dynamic pressure at each half-second interval. Finally, the code uses Matplotlib to graph the pressure curves, automatically identifying and annotating the peak maximum dynamic pressure (Max Q) for each of the three flight paths.